The Energy of Memorization: Why Graph-Level Label Noise Sharpens Representations and How to Stop It

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Graph Neural Networks (GNNs) are vulnerable to label noise in graph classification, yet the mechanisms driving this vulnerability are poorly understood. We empirically identify a property of the *learned representations* that tracks noise memorization: the within-graph Dirichlet energy of node features. Energy decreases as the model captures genuine class structure, then rises sharply, mainly in high-frequency components, when the model begins to memorize corrupted graph labels. We support this observation with a Poincaré-based lower bound: under a lower-Lipschitz regularity assumption on the encoder, fitting noisy graph-level labels for two structurally similar graphs forces a lower bound on their within-graph Dirichlet energies. Guided by this, suppressing the within-graph energy growth should block the noise-fitting path. We design and evaluate three interventions: a Dirichlet-energy regularizer ($\mathcal{L}_{DE}$), a robust loss (GCOD), and a spectral weight constraint (CE+W2). Across eight benchmarks, including the full and modified ogbg-ppa, symmetric noise up to 60%, asymmetric noise, and a transfer to Cora node classification, the three interventions all reduce within-graph energy and improve robustness while preserving clean-data performance. GCOD, our practical recommendation, matches or exceeds OMG [Yin et al., 2023] at lower training overhead.

1 Introduction

Graph Neural Networks (GNNs) are now the workhorse of *graph classification*, the task of assigning a single label to an entire graph, with applications spanning molecular property prediction [Wale and Karypis, 2006], bioinformatics [Borgwardt et al., 2005], and social-network analysis [Yanardag and Vishwanathan, 2015a]. In real deployments, the labels attached to graphs are routinely noisy: assays disagree, annotators err, and weak supervision injects systematic mistakes. Although learning under label noise has been thoroughly studied for images [Algan and Ulusoy, 2021] and for node classification [Dai et al., 2021, Yuan et al., 2023a], graph classification under noisy labels is dramatically less explored [NT et al., 2019, Yin et al., 2023].

Graph classification is a fundamentally different setting. In node classification, label noise corrupts *individual node labels* on a single graph, and homophily, the tendency of connected nodes to share labels, is a structural prior that directly couples Dirichlet energy of the *label signal* to noise. In graph classification, label noise flips the label of an *entire graph*: nodes have no individual classification labels, and within-graph homophily/heterophily is largely orthogonal to the graph-level label. The standard tools and intuitions from node classification therefore do not transfer.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Our central observation. We measure the Dirichlet energy of *learned node representations*, not of labels. Empirically, this energy first decreases as the model captures genuine, transferable patterns; then, when the model begins to memorize incorrect graph-level labels, it rises sharply, with the surge concentrated in the high-frequency spectrum. This rise is not a tautology: noise is injected only at the graph-label level, while node features and edges are unchanged, so any correlation between energy and labels has to emerge through training dynamics. Theorem 1 supports this observation analytically: under a lower-Lipschitz regularity assumption on the encoder, fitting noisy graph-level labels for two structurally similar graphs forces a lower bound on their within-graph Dirichlet energies via the Graph Poincaré inequality.

Why does suppressing this energy growth help? Theorem 1 shows that fitting graph-level label noise forces within-graph Dirichlet energy to increase. Suppressing this growth therefore plausibly blocks the noise-fitting path, a hypothesis we test with three mechanistically distinct interventions and validate empirically across eight benchmarks.

Contributions.

- A controlled study of *when* GNNs fit graph-level label noise (Section 4), isolating three failure modes: over-parameterization, low-order graphs, and small training sets.
- Within-graph Dirichlet energy as a diagnostic for graph-level noise memorization (Section 5), with a frequency decomposition that localizes the surge to high-frequency components and an empirical scaling with the noise rate.
- Analytical support for the diagnostic (Section 6): a spectral identity (Proposition 1) and a Poincaré-based lower bound under noise fitting (Theorem 1). The bound is unique to graph classification, where labels and feature smoothness live on different objects, unlike node classification.
- Three interventions (Section 7): a Dirichlet-energy regularizer $\mathcal{L}_{DE}$, a centroid-based loss GCOD, and a spectral weight constraint CE+W2. They are mechanistically distinct and all reduce within-graph energy while improving robustness; GCOD is our practical recommendation.
- Evaluation on eight benchmarks under symmetric, asymmetric, and high-rate (60%) noise, the full and a modified ogbg-ppa, and a transfer to Cora node classification. GCOD matches or exceeds OMG [Yin et al., 2023] and SOP [Liu et al., 2022] at $\sim 70\times$ lower compute.

2 Related Work

Learning under label noise. Robust loss functions [Ghosh et al., 2017, Patrini et al., 2017], sample re-weighting [Liu and Tao, 2015, Liu et al., 2022], early-learning regularizers [Arpit et al., 2017], and small-loss filtering [Gui et al., 2021] dominate the image-classification literature. Wani et al. [2023] introduce NCOD, a centroid-based outlier-discounting loss that we adapt to graphs.

Noisy-label learning on graphs. Most prior work concerns *node* classification, exploiting homophily, contrastive losses, or pseudo-labels [Dai et al., 2021, Yuan et al., 2023a,b, Li et al., 2024, Kang et al., 2018]. For *graph* classification, NT et al. [2019] propose a surrogate loss that discards estimated-noisy labels under specific assumptions, and OMG [Yin et al., 2023] combines contrastive learning, MixUp [Lim et al., 2022], and curriculum filtering. We differ in two ways: (i) we provide a spectral/energy-based explanation of *why* GNNs fit graph-level noise, rather than only mitigating it, and (ii) our practical method (GCOD) achieves better or comparable robustness at a fraction of the compute.

Dirichlet energy and over-smoothing. Dirichlet energy quantifies signal smoothness on graphs [Zhou and Schölkopf, 2005]. Existing analyses establish that GNNs act as low-pass filters whose energy decays exponentially with depth under spectral conditions [NT and Maehara, 2019, Cai and Wang, 2020, Oono and Suzuki, 2020, Rusch et al., 2023], and that weight-matrix eigenvalues control attraction/repulsion of node features [Giovanni et al., 2023]. Energy-based regularizers [Zhou et al., 2021, Bo et al., 2021, Chen et al., 2023] address *too little* energy (over-smoothing) on *node* classification. We address the opposite regime: *too much* within-graph energy caused by graph-level label noise. Qian et al. [2025] recently study over-smoothing for graph classification through a structural-entropy lens; an explicit quantitative connection between structural entropy and noise-driven sharpening is left as an open direction. A more comprehensive related-work discussion is in Appendix A.

3 Background

Notation. A graph is $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ with $N = |\mathcal{V}|$, degree matrix $\mathbf{D}$, adjacency $\mathbf{A} \in \{0,1\}^{N\times N}$, feature matrix $\mathbf{X} \in \mathbb{R}^{N\times m}$, combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ (with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$) and normalized Laplacian $\boldsymbol{\Delta} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$. We write $\lambda_2(\mathcal{G})$ for $\lambda_2(\mathbf{L})$ throughout and use $\boldsymbol{\Delta}$ only when explicitly stated. In *graph classification*, the dataset is $\mathcal{D} = \{(\mathcal{G}^i, \mathbf{y}_i)\}_{i=1}^n$, $\mathbf{y}_i \in \{0,1\}^{|C|}$ the one-hot graph label; we write c_i for the class. Under message passing, $\mathbf{H}_i^{l+1} = \mathrm{UP}_{\boldsymbol{\Omega}_l}(\mathbf{H}_i^l, \mathrm{AGGR}_{\mathbf{W}_l}(\mathbf{H}_i^l, \mathbf{A}^i))$, with $\mathbf{H}_i^0 \equiv \mathbf{X}^i$ and $\mathbf{Z}^i \equiv \mathbf{H}_i^L$. A permutation-invariant readout $f_\theta : \mathbb{R}^{N^i\times m'} \to \mathbb{R}^{|C|}$ maps node features to class probabilities $\hat{\mathbf{y}}_i$. Under noisy labels, the observed c_i may differ from the ground truth; the test set is clean.

Dirichlet energy on graphs. For graph $\mathcal{G}^i$ with representation $\mathbf{Z}^i \in \mathbb{R}^{N^i\times m'}$ we use the combinatorial Dirichlet energy

$$E^{\text{dir}}(\mathbf{Z}^i) = \sum_{(u,v)\in\mathcal{E}^i} \left\|\mathbf{Z}_u^i - \mathbf{Z}_v^i\right\|_2^2 = \operatorname{tr}\big((\mathbf{Z}^i)^\top \mathbf{L}^i \mathbf{Z}^i\big). \tag{1}$$

Small E^{dir} means smooth (low-frequency) representations within the graph; large E^{dir} means sharpening (high-frequency). The symmetrically-normalized variant $E^{\text{dir}}_{\text{sym}}(\mathbf{Z}) = \sum_{(u,v)} \|\mathbf{Z}_u/\sqrt{d_u} - \mathbf{Z}_v/\sqrt{d_v}\|_2^2 = \operatorname{tr}(\mathbf{Z}^\top\boldsymbol{\Delta}\mathbf{Z})$, used in some over-smoothing analyses [NT and Maehara, 2019, Oono and Suzuki, 2020], is equivalent up to degree-bounded constants and our qualitative claims hold for either; we use (1) because it pairs cleanly with the Poincaré inequality of Section 6. To track behavior across a class-restricted subset $\mathcal{S}$ of training graphs we use

$$E^{\text{dir}}_{\text{set}}(\mathcal{S}) = \frac{1}{|\mathcal{S}|}\sum_{\mathcal{G}^i\in\mathcal{S}} E^{\text{dir}}(\mathbf{Z}^i). \tag{2}$$

Crucially, (1) is computed on *learned representations*, not on labels. Within-graph homophily of features therefore evolves with training, decoupled from the graph-level label of $\mathcal{G}^i$.

4 When GNNs Fit Graph-Label Noise

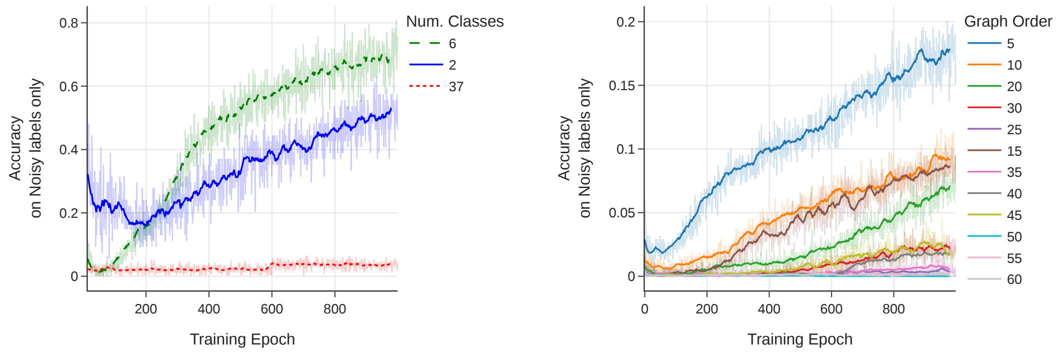
GNNs trained with standard cross-entropy (CE) on graph classification tasks are not uniformly vulnerable to label noise. On the full ogbg-ppa [Szklarczyk et al., 2018a] (37 classes, the standard OGB benchmark), even 40% symmetric noise barely degrades CE accuracy, because the model is under-parameterized for the full task and behaves robustly [Zhang et al., 2021]. To expose the noise-fitting regime, we control three axes: **(F1)** number of classes (proxy for over-parameterization), **(F2)** graph order (size of each graph), and **(F3)** training-set size. We use a 5-layer GIN [Xu et al., 2019] with 300 hidden units throughout, unless stated.

In all three regimes (Fig. 1 and Appendix D), GNNs progressively memorize noisy graph-level labels. The natural next question is therefore: what changes in the network's internal representations during memorization, and can this change be detected and controlled? The next section identifies Dirichlet energy as that internal signal.

5 Dirichlet Energy as a Diagnostic for Noise Fitting

We track $E^{\text{dir}}_{\text{set}}(\mathcal{D}_c)$, the within-graph energy averaged over training graphs of class c. Across architectures and datasets (Fig. 2), the diagnostic is consistent: $E^{\text{dir}}_{\text{set}}$ falls during the early "learning" phase, then rises precisely when the model begins to fit noisy graphs (CE-20% in Fig. 2b), with 40% noise yielding a strictly larger end-of-training energy than 10% noise (Appendix F).

Energy is sensitive to noise specifically through high-frequency components. A natural concern is that energy may also rises during clean-data overfitting [Yan et al., 2022]; we confirm this in Appendix F but show the noise-driven surge is markedly steeper. To localize the surge spectrally, we use the HLFF-GNN decomposition [Xu et al., 2024] to split representations into low-frequency (Z_1) and high-frequency (Z_2) components. On ENZYMES with 30% noise, $E^{\text{dir}}(Z_1)$ remains essentially unchanged, while $E^{\text{dir}}(Z_2)$ rises monotonically, and slope/variance statistics of $E^{\text{dir}}(Z_2)$ provide an



(a) Effect of class count.

(b) Effect of graph order.

Figure 1: Training accuracy on *noisy labels only*: a higher curve means more memorization. (a) Reducing the number of PPA classes (with 20% symmetric noise) makes a fixed GIN over-parameterized, accelerating noise fitting. (b) Smaller average graph order (synthetic data, 35% noise) leaves less internal structure for averaging, enabling memorization.

early warning for label noise (Appendix E). HLFF-GNN is used purely as an analytical tool; the same total-energy surge appears in standard GIN/GCN.

Why energy is more useful than a generic overfitting signal. Standard noise-detection signals require a *clean* held-out set: validation loss is uninformative when validation labels follow the same noise distribution as training (the noisy validation loss continues to decrease as the model fits noise, as we observe in our setting); train–test gap is unobservable at training time; small-loss heuristics need a calibration regime. Within-graph Dirichlet energy is computed entirely on the training graphs and their representations, so it remains a valid noise-fitting indicator under fully-noisy supervision. Beyond replacing the missing clean signal, it tells you *how* the network is failing: which spectral components drive the surge. This mechanistic specificity is what turns a diagnostic into a design principle, and motivates the three interventions in Section 7, each of which targets the same harmful spectral behavior in a distinct way.

6 Analytical Support: Why Energy Suppression Helps

Section 5 established Dirichlet energy as a reliable empirical signal. We now show this is not just correlation: under standard regularity, within-graph energy is *provably forced* to grow when a GNN fits graph-level label noise. The argument combines three pieces: a dataset-level spectral identity (Proposition 1), the Graph Poincaré inequality (Lemma 1), and a lower-Lipschitz bound on the GNN (Lemma 2) [Chuang and Jegelka, 2022, Davidson and Dym, 2025, Juvina et al., 2024], yielding Theorem 1.

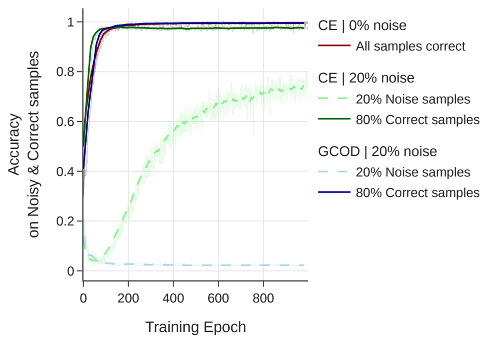
6.1 Three Building Blocks

Proposition 1 (Dataset-level Dirichlet energy)**.** *Let* $\mathcal{D} = \{\mathcal{G}^1, \ldots, \mathcal{G}^n\}$ *with representations* $\mathbf{Z}^i \in \mathbb{R}^{N^i \times m'}$ *and adjacencies* $\mathbf{A}^i$*. Define the block-diagonal graph* $\bar{\mathcal{G}} = (\bar{\mathbf{Z}}, \bar{\mathbf{A}})$ *with* $\bar{\mathbf{Z}} = [\mathbf{Z}^1 \| \ldots \| \mathbf{Z}^n] \in \mathbb{R}^{N_{tot} \times m'}$ *(vertical concatenation) and* $\bar{\mathbf{A}} = \mathrm{blkdiag}(\mathbf{A}^1, \ldots, \mathbf{A}^n)$*, with associated combinatorial Laplacian* $\bar{\mathbf{L}} = \mathrm{blkdiag}(\mathbf{L}^1, \ldots, \mathbf{L}^n)$*. Then*

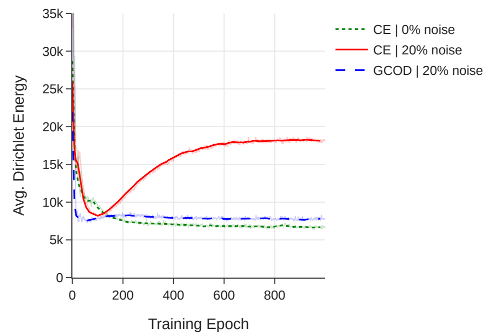
$$E^{dir}_{set}(\mathcal{D}) = \tfrac{1}{|\mathcal{D}|} E^{dir}(\bar{\mathbf{Z}}) = \tfrac{1}{|\mathcal{D}|} \sum_{r=1}^{m'} \sum_{u=1}^{N_{tot}} \bar{\lambda}_u (\bar{\boldsymbol{\psi}}_u^\top \bar{\mathbf{Z}}_{:,r})^2, \tag{3}$$

where $\{(\bar{\lambda}_u, \bar{\boldsymbol{\psi}}_u)\}$ *are eigenpairs of* $\bar{\mathbf{L}}$*, and the spectrum decomposes as* $\bar{\Lambda} = \bigcup_i \Lambda^i$ *with each eigenvector supported on a single connected component* $\mathcal{G}^i$ *(graph or block).*

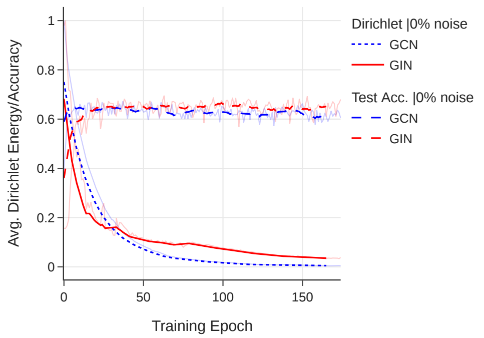
The proof (Appendix B) follows from the block-diagonal structure of $\bar{\mathbf{L}}$. The key structural fact: *each high-frequency eigenvector of* $\bar{\mathbf{L}}$ *is owned by exactly one graph*, so sharpening along it raises only that graph's energy.



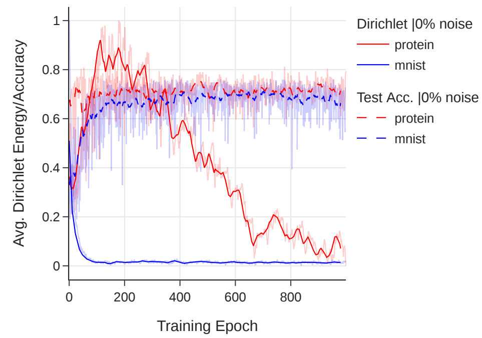
(a) Train accuracy (PPA-30%-6).



(b) Within-graph $E_{\text{set}}^{\text{dir}}$ (PPA-30%-6).



(c) Energy/accuracy across architectures.



(d) Across datasets.

Figure 2: Within-graph Dirichlet energy of learned representations *rises during noise memorization*, across architectures (GIN/GCN) and datasets (PPA, MUTAG, MNIST, PROTEINS). Under clean labels, $E_{\text{set}}^{\text{dir}}$ steadily decays as accuracy improves; under noise, it follows the same decay until the model starts fitting noisy graphs, after which it sharply rises. GCOD suppresses this surge.

Lemma 1 (Graph Poincaré, vector-valued form)**.** *For any feature matrix $\mathbf{Z} \in \mathbb{R}^{N\times m'}$ on a connected graph $\mathcal{G}$ with combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ and second-smallest eigenvalue $\lambda_2(\mathcal{G}) > 0$, let $\bar{\mathbf{z}} = \frac{1}{N}\sum_v \mathbf{Z}_v$. Then*

$$\sum_{v=1}^{N} \|\mathbf{Z}_v - \bar{\mathbf{z}}\|_2^2 \;\leq\; \frac{1}{\lambda_2(\mathcal{G})}\, E^{dir}(\mathbf{Z}). \tag{4}$$

(4) bounds within-graph feature variance by Dirichlet energy scaled by the spectral gap; proof by Courant–Fischer per feature dimension (Appendix B).

Lemma 2 (Lower Lipschitz bound for a GIN block with 2-layer combine, adapted from Chuang and Jegelka, 2022, Davidson and Dym, 2025, Juvina et al., 2024)**.** *Let $h_\theta(\mathbf{X}) = \sigma_2\big(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2\big)$ be a single GIN aggregation $\mathbf{M} = (1+\varepsilon)\mathbf{I} + \mathbf{A} \in \mathbb{R}^{N\times N}$ followed by a two-layer MLP with weights $\mathbf{W}_1 \in \mathbb{R}^{m\times m'}$ and $\mathbf{W}_2 \in \mathbb{R}^{m'\times m''}$ acting on the feature dimension, with the widening regime $m \leq m' \leq m''$ and full row rank weights. Assume $\sigma_{\min}(\mathbf{M}) \geq \kappa > 0$ (e.g., $1+\varepsilon > |\lambda_{\min}(\mathbf{A})|$), $\sigma_{\min}(\mathbf{W}_k) \geq m_k > 0$, and that σ_1, σ_2 are leaky-ReLU with slope $\alpha > 0$. Then*

$$\|h_\theta(\mathbf{X} + \Delta\mathbf{X}) - h_\theta(\mathbf{X})\|_F \;\geq\; c_{lower}\, \|\Delta\mathbf{X}\|_F, \qquad c_{lower} = \alpha^2 m_1 m_2 \kappa. \tag{5}$$

For ordinary ReLU the chain applies locally with $\alpha = 1$ on segments where all preactivations are strictly positive, giving $c_{lower} = m_1 m_2 \kappa$. A true two-layer GIN $\sigma_2(\mathbf{M}\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2)$ replaces c_{lower} by $\alpha^2 m_1 m_2 \kappa^2$.

Proof: chain the global element-wise lower-Lipschitz bound of leaky-ReLU ($\|\sigma_k(\mathbf{Y} + \Delta) - \sigma_k(\mathbf{Y})\|_F \geq \alpha\|\Delta\|_F$) with $\sigma_{\min}$ of $\mathbf{W}_2, \mathbf{W}_1, \mathbf{M}$ via $\sigma_{\min}(\mathbf{P} \otimes \mathbf{Q}) = \sigma_{\min}(\mathbf{P})\sigma_{\min}(\mathbf{Q})$ (Appendix B). The lemma encodes that the GIN does not collapse meaningful feature differences under injective aggregation and feature transforms, a property the model can exploit when fitting noisy labels.

6.2 Main Theorem: Noise Fitting Forces Energy Growth

Consider two graphs $\mathcal{G}^i, \mathcal{G}^j$ with $N^i = N^j = N$ and similar pooled descriptors $\mathbf{g}_i \approx \mathbf{g}_j$ on the clean distribution (same-N for notational simplicity; unequal N in Appendix B). Suppose label noise flips c_j to $c'_i \neq c_i$, forcing different graph-level outputs. Recall $\mathbf{Z}^i \in \mathbb{R}^{N\times m'}$ from Section 3; let $\mathbf{g}_i = \frac{1}{N}\sum_v \mathbf{Z}^i_v$ and $\widetilde{\mathbf{Z}}^i := \mathbf{1}_N \mathbf{g}_i^\top$ (constant-signal replication of $\mathbf{g}_i$).

Theorem 1 (Graph-level noise fitting forces E^{dir} to grow)**.** *Under the hypotheses of Lemmas 1–2 and $N^i = N^j = N$,*

$$\|\mathbf{Z}^i - \mathbf{Z}^j\|_F \;\leq\; \sqrt{N}\,\|\mathbf{g}_i - \mathbf{g}_j\|_2 + \sqrt{\tfrac{E^{\mathit{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\tfrac{E^{\mathit{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}}. \tag{6}$$

In particular, if the noisy training objective combined with an L-Lipschitz readout forces a graph-level output gap of at least Δ_{out}, so that $\|\mathbf{Z}^i - \mathbf{Z}^j\|_F \geq \Theta_{noise} := L^{-1}\Delta_{out}$, while the structural similarity gives $\|\mathbf{g}_i - \mathbf{g}_j\|_2 \leq \varepsilon$, then

$$\sqrt{\tfrac{E^{\mathit{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\tfrac{E^{\mathit{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}} \;\geq\; \Theta_{noise} - \sqrt{N}\,\varepsilon. \tag{7}$$

Reading the bound. (7) lower-bounds the within-graph Dirichlet energies. The intuition is geometric: for two structurally similar graphs mapped to different labels, $\mathbf{g}_i \approx \mathbf{g}_j$, so the required output gap Θ_{noise} cannot come from the means alone; by Poincaré, the only remaining route is the within-graph variance E^{dir}/λ_2. The unequal-N case (zero-padded matching) is handled in Appendix B. The bound has no node-classification analog: there labels live on the same nodes whose smoothness E^{dir} measures, so any link is trivial by construction.

Empirical anchor. The bound predicts that $\sqrt{E^{\text{dir}}/\lambda_2}$ should grow monotonically with the noise rate, since higher noise forces the encoder to discriminate more pairs of similar graphs. The PPA noise-rate sweep in Appendix F confirms this: asymptotic $E^{\text{dir}}_{\text{set}}$ at $10\%, 20\%, 30\%, 40\%$ noise is monotonically increasing.

Implication. Theorem 1 establishes that fitting graph-level label noise forces within-graph Dirichlet energy to increase. Suppressing this growth therefore blocks the noise-fitting path; we validate this empirically in Section 8. An explicit quantitative connection between which graphs absorb most of the energy growth and structural properties such as Shannon entropy [Qian et al., 2025] remains an open question.

7 Three Methods that Reduce Within-Graph Energy

The diagnostic and analytical results above suggest controlling within-graph Dirichlet energy through any of three mechanistically distinct levers: a regularizer on the energy itself, a loss that down-weights samples driving the energy up, or an architectural projection that removes the spectral directions responsible for sharpening. We instantiate one intervention per lever. GCOD is our practical recommendation; CE+W2 is included as a theoretical instantiation.

7.1 Explicit Dirichlet-Energy Regularization ($\mathcal{L}_{DE}$)

The most direct lever is to penalize $E^{\text{dir}}(\mathbf{Z}^i)$ above a class-aware threshold U_{c_i}:

$$\mathcal{L}_{DE}(\mathcal{D}) = \frac{1}{N}\sum_{i=1}^{N}\big[\max(0, E^{\text{dir}}(\mathbf{Z}^i) - U_{c_i})\big]^2, \quad \mathcal{L} = \mathcal{L}_{CE} + \lambda\,\mathcal{L}_{DE}. \tag{8}$$

We instantiate the threshold in two ways. **Class-specific:** U_c is recomputed every epoch as the mean E^{dir} over a small clean validation subset of class c. This adapts to class-dependent baseline energies (a molecular graph's natural energy differs from a social graph's), but requires clean held-out data. **Fixed:** $U_c = U$ globally, a single hyperparameter, no clean validation needed, at the cost of mild over-smoothing on clean labels. Both variants improve robustness; class-specific additionally improves *clean-label* accuracy by acting as a generalization-aware regularizer.

7.2 Graph Centroid Outlier Discounting (GCOD)

GCOD is our practical recommendation. It extends NCOD [Wani et al., 2023] (originally an image-classification loss) to graph classification, with two design changes that are necessary in the graph-classification regime: a third KL term that disentangles clean from noisy graphs via per-sample trainable parameters $\mathbf{u} \in [0,1]^n$, and feedback through the current training accuracy a_{train} that prevents premature outlier discounting. For batch B with predictions $f_\theta(\mathbf{Z}_B)$, hard predictions $\hat{\mathbf{y}}_B$, soft labels $\tilde{\mathbf{y}}_B$ (centroid-based, see Appendix C.6), and observed labels $\mathbf{y}_B$:

$$\mathcal{L}_1 = \mathcal{L}_{CE}\big(f_\theta(\mathbf{Z}_B) + a_{\text{train}}\, \text{diag}_{\text{vec}}(\mathbf{u}_B)\, \mathbf{y}_B,\ \tilde{\mathbf{y}}_B\big), \tag{9}$$

$$\mathcal{L}_2 = \frac{1}{|C|}\big\|\hat{\mathbf{y}}_B + \text{diag}_{\text{vec}}(\mathbf{u}_B)\, \mathbf{y}_B - \mathbf{y}_B\big\|^2, \tag{10}$$

$$\mathcal{L}_3 = (1 - a_{\text{train}})\, D_{KL}\big\{\mathfrak{L},\ \sigma(-\log \mathbf{u}_B)\big\}, \tag{11}$$

with $\mathfrak{L} = \log \sigma(\text{diag}_{\text{mat}}(f_\theta(\mathbf{Z}_B)\mathbf{y}_B^\top))$. Updates are split: $\theta \leftarrow \theta - \alpha \nabla_\theta(\mathcal{L}_1 + \mathcal{L}_3)$, $\mathbf{u} \leftarrow \mathbf{u} - \beta \nabla_{\mathbf{u}} \mathcal{L}_2$. GCOD is not a theoretically derived energy minimizer. Empirically, however, it reduces $E^{\text{dir}}_{\text{set}}$ during training (Fig. 2) without explicitly penalizing it: by down-weighting graphs flagged as noisy via large u_i, it removes the gradient pressure that would otherwise push the network toward high-frequency directions, exactly as Theorem 1 predicts for any sample-level mechanism that suppresses noise-driven gradients.

7.3 Spectral Constraint on Weight Matrices (CE+W2)

A third lever, included as a theoretical instantiation, acts directly on the weights. Giovanni et al. [2023] show that for linear graph convolutions with symmetric $\mathbf{W}$, the gradient flow of E^{dir} decomposes along eigenvectors of $\mathbf{W}$: positive eigenvalues smooth, negative ones sharpen. Practical $\mathbf{W}_l^2$ is not symmetric, so we split $\mathbf{W}_l^2 = \mathbf{S}_l + \mathbf{N}_l$ and project only $\mathbf{S}_l$ to its positive eigenspace, leaving the skew $\mathbf{N}_l$ (purely imaginary spectrum, rotational) untouched. This is a heuristic extension of GRAFF — the energy of $\mathbf{MZW}$ depends on $\mathbf{W}^\top\mathbf{W}$, not just $\mathbf{S}_l$ — but the resulting CE+W2 needs neither clean validation nor extra hyperparameters and shows the energy mechanism can be controlled architecturally. Implementation details are in Appendix C.4; the per-epoch eigendecomposition is costly and can destabilize convergence, so we recommend $\mathcal{L}_{DE}$ or GCOD for practical use.

8 Experiments

Setup. We evaluate on ogbg-ppa [Szklarczyk et al., 2018a] (full and a controlled 30%/6-class subset), MUTAG, PROTEINS, ENZYMES, IMDB-BINARY, REDDIT-MULTI-12K, MSRC-21, MNIST-graph, and Cora (node classification, transfer test).[1] Backbones are GIN [Xu et al., 2019] and GCN [Kipf and Welling, 2017]; full hyperparameters are in Appendix C. We compare against three baselines: standard cross-entropy (CE); SOP [Liu et al., 2022], a strong noise-robust loss from image classification; and OMG [Yin et al., 2023], the leading graph-classification noise method. All numbers are mean±std over 4 runs.

Why a controlled PPA subset? On the full 37-class PPA, all methods are under-parameterized for the task, and CE itself is robust (Section 4); a noise-robustness method cannot help where the baseline does not fail. We use a 30%/6-class PPA subset as a *controlled* testbed where over-parameterization triggers noise fitting, and we additionally report the full-PPA result to show our methods do no harm there. Generality claims are grounded in the seven additional benchmarks.

8.1 Main Result: Controlled PPA Subset

Table 1 reports the full ablation on the controlled PPA subset. GCOD is best at every noise level and, crucially, has the smallest gap between best and final test accuracy, indicating that it does not overfit late in training. CE+W2 and $\mathcal{L}_{DE}$ both improve robustness over CE, confirming the convergent-evidence prediction of Section 6. Under clean labels, $\mathcal{L}^*_{DE}$ even outperforms CE, a direct effect of the class-aware energy threshold.

[1] Anonymized code: `https://anonymous.4open.science/r/Robustness_Graph_Classification-E76F/`.

Table 1: Test accuracy on the PPA-30%-6 subset (5-layer GIN, mean±std over 4 runs). **Best** is highlighted in red and bold, second-best in blue. "CE clean only" shows accuracy on the clean subset of the noisy data, as a reference upper bound. Class-specific $\mathcal{L}^*_{DE}$ requires a clean validation set.

Noise	**Method**	**Test Acc.**		**Train Acc.**	
		Best	Last	Best	Last
0%	CE	96.25 ± 0.05	91.25	100.0	99.33
	GCOD	96.65 ± 0.52	93.25	99.23	99.03
	CE+W2	96.50 ± 1.12	86.50	99.76	99.29
	$\mathcal{L}_{DE}$ (Fixed)	96.58 ± 0.27	85.70	99.26	98.29
	$\mathcal{L}^*_{DE}$ (Class-spec.)	96.96 ± 0.04	88.67	99.41	98.67
20%	CE (clean only)	96.15 ± 0.09	90.66	84.50	84.07
	CE	88.66 ± 0.16	62.58	94.98	93.45
	SOP	91.01 ± 0.44	85.50	79.17	77.64
	GCOD	93.91 ± 0.26	92.58	80.11	79.52
	CE+W2	89.83 ± 1.23	76.66	84.90	83.68
	$\mathcal{L}_{DE}$ (Fixed)	89.69 ± 0.57	80.25	78.07	70.14
	$\mathcal{L}^*_{DE}$	92.34 ± 0.41	80.92	84.55	83.73
40%	CE (clean only)	95.08 ± 0.13	80.44	68.15	67.05
	CE	82.08 ± 0.22	51.83	82.47	78.88
	SOP	82.33 ± 1.32	65.41	58.90	57.68
	GCOD	93.88 ± 0.04	91.08	65.09	64.31
	CE+W2	88.25 ± 1.13	56.33	68.78	65.88
	$\mathcal{L}_{DE}$ (Fixed)	88.58 ± 0.75	77.12	61.08	54.53
	$\mathcal{L}^*_{DE}$	88.55 ± 0.11	71.83	68.98	67.80

Table 2: GCOD matches or beats CE across diverse datasets. Mean over 4 runs; ±std reported where available. *Sym*$_{20}$*:* 20% symmetric noise. *Asym*$_{40}$*:* 40% asymmetric noise. The Asym$_{40}$ GCOD column reports the std deviations from [Wani et al., 2023]'s rebuttal protocol.

Dataset	**0% CE**	**Sym$_{20}$ CE**	**Sym$_{20}$ GCOD**	**Asym$_{40}$ CE**	**Asym$_{40}$ GCOD**
PROTEINS	81.16	76.23	79.38	72.90	76.19 ± 1.22
MNIST	72.69	66.30	71.26	71.15	72.61 ± 0.43
ENZYMES	73.33	64.16	68.69	65.80	69.81 ± 0.98
IMDB-B	76.50	75.00	75.40	68.18	72.89 ± 0.91
MUTAG	94.73	89.47	90.01	91.16	93.19 ± 0.38
REDDIT	48.15	45.05	44.98	48.01	47.94 ± 2.87
MSRC-21	96.69	90.26	94.69	94.39	95.57 ± 0.32

8.2 Generalization Across Datasets and Noise Types

Table 2 reports GCOD across seven benchmarks beyond PPA. GCOD consistently improves over CE under both 20% symmetric and 40% asymmetric noise, and matches CE on REDDIT where the baseline is already saturated by class imbalance. The empirical trend that gains grow with the noise rate (Appendix F) is consistent with Theorem 1: the forced energy lower bound $\Theta_{\text{noise}} - \sqrt{N}\varepsilon$ scales with the noise level, so the room a method has to help also scales.

Full ogbg-ppa. Under the full 37-class benchmark, all methods including CE behave robustly to noise (Section 4); we therefore use it only as a do-no-harm check. Detailed results are in Appendix H.

Table 3: OMG protocol (Yin et al., 2023). Percentage gain over CE.

Dataset	OMG	GCOD
MUTAG	0.061	0.062
IMDB-B	0.047	0.049
PROTEINS	0.039	0.041

Table 4: Relative training time (CE-GIN = 1.00).

Method	Runtime
CE-GIN	1.00
SOP	1.048
GCOD	**1.029**
CE+W2	1.33
OMG [Yin et al., 2023]	≈ 3.00

Comparison with OMG. Under the OMG [Yin et al., 2023] protocol (Table 3), GCOD matches OMG's accuracy (within 0.1–0.2 pp) while requiring $\sim$70$\times$ less training overhead (Table 4): OMG adds $\approx$200% runtime, GCOD only $\approx$2.9%.

Cross-setting transfer to node classification (Cora). Applied unchanged to Cora under three structured 20% noise types (flip, random, uniform) against 13 node-specific baselines (full table in Appendix I), at least one of our methods reaches the top four on every backbone-noise combination, with several top-three placements. Highlights: on GAT, GCOD reaches 0.787 (#2) under uniform noise and CE+W2 0.8157 (#2) under clean labels; on GCN, CE+W2 reaches 0.7743 (#2) under random noise and GCOD 0.7687 (#3) under flip noise. That an energy-based regularizer using neither neighbor labels nor homophily matches specialized node-classification methods is direct evidence the perspective captures a general property of GNN training under label noise.

Convergent evidence. The three methods (architectural, regularization, sample re-weighting) all reduce $E_{\text{set}}^{\text{dir}}$ during training and improve robustness over CE — convergent evidence for the energy mechanism predicted by Theorem 1 (Fig. 2, Appendix F).

9 Conclusion

Within-graph Dirichlet energy diagnoses graph-level noise memorization, supported by a Poincaré-based lower bound (Theorem 1) unique to graph classification. Three interventions ($\mathcal{L}_{DE}$, GCOD, CE+W2) reduce energy and improve robustness on eight benchmarks; GCOD matches OMG at $\sim$70$\times$ lower compute. **Limitations:** sufficient bound only, synthetic noise, heuristic CE+W2. **Impact:** robust GNNs aid chemistry/biology; GCOD needs no clean validation.

References

Görkem Algan and Ilkay Ulusoy. Image classification with deep learning in the presence of noisy labels: A survey. *Knowledge-Based Systems*, 215:106771, 2021. ISSN 0950-7051. doi: https://doi.org/10.1016/j.knosys.2021.106771. URL `https://www.sciencedirect.com/science/article/pii/S0950705121000344`.

Eric Arazo, Diego Ortego, Paul Albert, Noel O'Connor, and Kevin Mcguinness. Unsupervised label noise modeling and loss correction. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 312–321. PMLR, 09–15 Jun 2019. URL `https://proceedings.mlr.press/v97/arazo19a.html`.

Devansh Arpit, Stanisław Jastrzębski, Nicolas Ballas, David Krueger, Emmanuel Bengio, Maxinder S. Kanwal, Tegan Maharaj, Asja Fischer, Aaron Courville, Yoshua Bengio, and Simon Lacoste-Julien. A closer look at memorization in deep networks. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 233–242. PMLR, 06–11 Aug 2017. URL `https://proceedings.mlr.press/v70/arpit17a.html`.

Deyu Bo, Xiao Wang, Chuan Shi, and Huawei Shen. Beyond low-frequency information in graph convolutional networks. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(5): 3950–3957, May 2021. doi: 10.1609/aaai.v35i5.16514. URL `https://ojs.aaai.org/index.php/AAAI/article/view/16514`.

Karsten M. Borgwardt, Cheng Soon Ong, Stefan Schönauer, S. V. N. Vishwanathan, Alex J. Smola, and Hans-Peter Kriegel. Protein function prediction via graph kernels. *Bioinformatics*, 21(suppl_1): i47–i56, 06 2005. ISSN 1367-4803. doi: 10.1093/bioinformatics/bti1007. URL https://doi.org/10.1093/bioinformatics/bti1007.

Chen Cai and Yusu Wang. A note on over-smoothing for graph neural networks, 2020. URL https://arxiv.org/abs/2006.13318.

Jialin Chen, Yuelin Wang, Cristian Bodnar, Rex Ying, Pietro Lio, and Yu Guang Wang. Dirichlet energy enhancement of graph neural networks by framelet augmentation, 2023. URL https://arxiv.org/abs/2311.05767.

Ching-Yao Chuang and Stefanie Jegelka. Tree mover's distance: Bridging graph metrics and stability of graph neural networks. *proceedings neurips*, 35:2944–2957, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/file/139ae969f49abd9a113981c1f7fce5ce-Paper-Conference.pdf.

Enyan Dai, Charu Aggarwal, and Suhang Wang. Nrgnn: Learning a label noise resistant graph neural network on sparsely and noisily labeled graphs. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, KDD '21, page 227–236, New York, NY, USA, 2021. Association for Computing Machinery. ISBN 9781450383325. doi: 10.1145/3447548.3467364. URL https://doi.org/10.1145/3447548.3467364.

Enyan Dai, Wei Jin, Hui Liu, and Suhang Wang. Towards robust graph neural networks for noisy graphs with sparse labels. In *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*, WSDM '22, page 181–191, New York, NY, USA, 2022. Association for Computing Machinery. ISBN 9781450391320. doi: 10.1145/3488560.3498408. URL https://doi.org/10.1145/3488560.3498408.

Yair Davidson and Nadav Dym. On the holder stability of multiset and graph neural networks. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=P7KIGdgW8S.

Xuefeng Du, Tian Bian, Yu Rong, Bo Han, Tongliang Liu, Tingyang Xu, Wenbing Huang, Yixuan Li, and Junzhou Huang. Noise-robust graph learning by estimating and leveraging pairwise interactions. *Transactions on Machine Learning Research*, 2023. ISSN 2835-8856. URL https://openreview.net/forum?id=r7imkFEAQb.

James Fox and Sivasankaran Rajamanickam. How robust are graph neural networks to structural noise? *ArXiv*, abs/1912.10206, 2019. URL https://api.semanticscholar.org/CorpusID:209444207.

Fernando Gama, Joan Bruna, and Alejandro Ribeiro. Stability properties of graph neural networks. *IEEE Transactions on Signal Processing*, 68:5680–5695, 2019. URL https://api.semanticscholar.org/CorpusID:152282530.

Aritra Ghosh, Himanshu Kumar, and P Shanti Sastry. Robust loss functions under label noise for deep neural networks. In *Proceedings of the AAAI conference on artificial intelligence*, volume 31, 2017.

Francesco Di Giovanni, James Rowbottom, Benjamin Paul Chamberlain, Thomas Markovich, and Michael M. Bronstein. Understanding convolution on graphs via energies. *Transactions on Machine Learning Research*, 2023. ISSN 2835-8856. URL https://openreview.net/forum?id=v5ew3FPTgb.

Xian-Jin Gui, Wei Wang, and Zhang-Hao Tian. Towards understanding deep learning from noisy labels with small-loss criterion. In Zhi-Hua Zhou, editor, *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence, IJCAI-21*, pages 2469–2475. International Joint Conferences on Artificial Intelligence Organization, 8 2021. doi: 10.24963/ijcai.2021/340. URL https://doi.org/10.24963/ijcai.2021/340. Main Track.

Bo Han, Quanming Yao, Xingrui Yu, Gang Niu, Miao Xu, Weihua Hu, Ivor Tsang, and Masashi Sugiyama. Co-teaching: Robust training of deep neural networks with extremely noisy labels. *Advances in neural information processing systems*, 31, 2018.

Simona Juvina, Ana-Antonia Neacșu, Jean-Christophe Pesquet, Burileanu Corneliu, Jérôme Rony, and Ismail Ben Ayed. Training graph neural networks subject to a tight lipschitz constraint. *Transactions on Machine Learning Research Journal*, 2024.

Zhao Kang, Haiqi Pan, Steven C. H. Hoi, and Zenglin Xu. Robust graph learning from noisy data. *IEEE Transactions on Cybernetics*, 50:1833–1843, 2018. URL `https://api.semanticscholar.org/CorpusID:56208557`.

Jihye Kim, Aristide Baratin, Yan Zhang, and Simon Lacoste-Julien. Crosssplit: Mitigating label noise memorization through data splitting. In *International Conference on Machine Learning*, pages 16377–16392. PMLR, 2023.

Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 2017. URL `https://openreview.net/forum?id=SJU4ayYgl`.

Nils M. Kriege and Petra Mutzel. Subgraph matching kernels for attributed graphs. In *International Conference on Machine Learning*, 2012. URL `https://api.semanticscholar.org/CorpusID:8790929`.

Junnan Li, Richard Socher, and Steven C. H. Hoi. Dividemix: Learning with noisy labels as semi-supervised learning. In *ICLR*, 2020. URL `https://openreview.net/forum?id=HJgExaVtwr`.

Xianxian Li, Qiyu Li, De Li, Haodong Qian, and Jinyan Wang. Contrastive learning of graphs under label noise. *Neural Networks*, 172:106113, 2024. ISSN 0893-6080. doi: https://doi.org/10.1016/j.neunet.2024.106113. URL `https://www.sciencedirect.com/science/article/pii/S0893608024000273`.

Soon Hoe Lim, N. Benjamin Erichson, Francisco Utrera, Winnie Xu, and Michael W. Mahoney. Noisy feature mixup. In *International Conference on Learning Representations*, 2022. URL `https://openreview.net/forum?id=vJb4I2ANmy`.

Sheng Liu, Zhihui Zhu, Qing Qu, and Chong You. Robust training under label noise by over-parameterization. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato, editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 14153–14172. PMLR, 17–23 Jul 2022. URL `https://proceedings.mlr.press/v162/liu22w.html`.

Tongliang Liu and Dacheng Tao. Classification with noisy labels by importance reweighting. *IEEE Transactions on pattern analysis and machine intelligence*, 38(3):447–461, 2015.

Marion Neumann, Roman Garnett, Christian Bauckhage, and Kristian Kersting. Propagation kernels: Efficient graph kernels from propagated information. *Machine Learning*, 102(2):209–245, 2016. ISSN 1573-0565. doi: 10.1007/s10994-015-5517-9. URL `https://doi.org/10.1007/s10994-015-5517-9`.

Hoang NT and Takanori Maehara. Revisiting graph neural networks: All we have is low-pass filters. *ArXiv*, abs/1905.09550, 2019. URL `https://api.semanticscholar.org/CorpusID:162183860`.

Hoang NT, Jun Jin Choong, and Tsuyoshi Murata. Learning graph neural networks with noisy labels, 2019. URL `https://openreview.net/forum?id=r1xOmNmxuN`.

Kenta Oono and Taiji Suzuki. Graph neural networks exponentially lose expressive power for node classification. In *International Conference on Learning Representations*, 2020. URL `https://openreview.net/forum?id=S1ldO2EFPr`.

Giorgio Patrini, Alessandro Rozza, Aditya Krishna Menon, Richard Nock, and Lizhen Qu. Making deep neural networks robust to label noise: A loss correction approach. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1944–1952, 2017.

Geoff Pleiss, Tianyi Zhang, Ethan Elenberg, and Kilian Q Weinberger. Identifying mislabeled data using the area under the margin ranking. *Advances in Neural Information Processing Systems*, 33: 17044–17056, 2020.

Feifei Qian, Lu Bai, Lixin Cui, Ming Li, Hangyuan Du, Yue Wang, and Edwin Hancock. Exploring the over-smoothing problem of graph neural networks for graph classification: An entropy-based viewpoint. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI)*, pages 3235–3243, 2025.

Siyi Qian, Haochao Ying, Renjun Hu, Jingbo Zhou, Jintai Chen, Danny Z. Chen, and Jian Wu. Robust training of graph neural networks via noise governance. In *Proceedings of the Sixteenth ACM International Conference on Web Search and Data Mining*, WSDM '23, page 607–615, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9781450394079. doi: 10.1145/3539597.3570369. URL `https://doi.org/10.1145/3539597.3570369`.

Scott E. Reed, Honglak Lee, Dragomir Anguelov, Christian Szegedy, Dumitru Erhan, and Andrew Rabinovich. Training deep neural networks on noisy labels with bootstrapping. In *ICLR 2015*, 2015. URL `http://arxiv.org/abs/1412.6596`.

T. Konstantin Rusch, Michael M. Bronstein, and Siddhartha Mishra. A survey on oversmoothing in graph neural networks. *ArXiv*, abs/2303.10993, 2023. URL `https://api.semanticscholar.org/CorpusID:257632346`.

Damian Szklarczyk, Annika Gable, David Lyon, Alexander Junge, Stefan Wyder, Jaime Huerta-Cepas, Milan Simonovic, Nadezhda Doncheva, John Morris, Peer Bork, Lars Jensen, and Christian von Mering. String v11: protein-protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic acids research*, 47, 11 2018a. doi: 10.1093/nar/gky1131.

Damian Szklarczyk, Annika L Gable, David Lyon, Alexander Junge, Stefan Wyder, Jaime Huerta-Cepas, Milan Simonovic, Nadezhda T Doncheva, John H Morris, Peer Bork, Lars J Jensen, and Christianvon Mering. STRING v11: protein–protein association networks with increased coverage, supporting functional discovery in genome-wide experimental datasets. *Nucleic Acids Research*, 47(D1):D607–D613, 11 2018b. ISSN 0305-1048. doi: 10.1093/nar/gky1131. URL `https://doi.org/10.1093/nar/gky1131`.

Nikil Wale and George Karypis. Comparison of descriptor spaces for chemical compound retrieval and classification. In *Sixth International Conference on Data Mining (ICDM'06)*, pages 678–689, 2006. doi: 10.1109/ICDM.2006.39.

Farooq Ahmad Wani, Maria Sofia Bucarelli, and Fabrizio Silvestri. Learning with noisy labels through learnable weighting and centroid similarity. *2024 International Joint Conference on Neural Networks (IJCNN)*, pages 1–9, 2023. URL `https://api.semanticscholar.org/CorpusID:257557810`.

Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. How powerful are graph neural networks? In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6-9, 2019*. OpenReview.net, 2019. URL `https://openreview.net/forum?id=ryGs6iA5Km`.

Shilong Xu, Jipeng Guo, Jiahao Long, Mingliang Cui, and Youqing Wang. High-low frequency filtering aware graph neural networks with representation decoupling learning. In *2024 China Automation Congress (CAC)*, pages 4005–4010, 2024. doi: 10.1109/CAC63892.2024.10864603.

Yujun Yan, Milad Hashemi, Kevin Swersky, Yaoqing Yang, and Danai Koutra. Two sides of the same coin: Heterophily and oversmoothing in graph convolutional neural networks. In *2022 IEEE International Conference on Data Mining (ICDM)*, pages 1287–1292, 2022. doi: 10.1109/ICDM54844.2022.00169.

Pinar Yanardag and S.V.N. Vishwanathan. Deep graph kernels. In *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '15, page 1365–1374, New York, NY, USA, 2015a. Association for Computing Machinery. ISBN 9781450336642. doi: 10.1145/2783258.2783417. URL `https://doi.org/10.1145/2783258.2783417`.

Pinar Yanardag and S.V.N. Vishwanathan. Deep graph kernels. In *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '15, page 1365–1374, New York, NY, USA, 2015b. Association for Computing Machinery. ISBN 9781450336642. doi: 10.1145/2783258.2783417. URL `https://doi.org/10.1145/2783258.2783417`.

Nan Yin, Li Shen, Mengzhu Wang, Xiao Luo, Zhigang Luo, and Dacheng Tao. Omg: Towards effective graph classification against label noise. *IEEE Transactions on Knowledge and Data Engineering*, 35(12):12873–12886, 2023. doi: 10.1109/TKDE.2023.3271677.

Jingyang Yuan, Xiao Luo, Yifang Qin, Yusheng Zhao, Wei Ju, and Ming Zhang. Learning on graphs under label noise. In *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, 2023a. doi: 10.1109/ICASSP49357.2023.10096088.

Jinliang Yuan, Hualei Yu, Meng Cao, Jianqing Song, Junyuan Xie, and Chongjun Wang. Self-supervised robust graph neural networks against noisy graphs and noisy labels. *Applied Intelligence*, 53(21):25154–25170, aug 2023b. ISSN 0924-669X. doi: 10.1007/s10489-023-04836-6. URL `https://doi.org/10.1007/s10489-023-04836-6`.

Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning (still) requires rethinking generalization. *Commun. ACM*, 64(3):107–115, February 2021. ISSN 0001-0782. doi: 10.1145/3446776. URL `https://doi.org/10.1145/3446776`.

Hongyi Zhang, Moustapha Ciss'e, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. *CoRR*, abs/1710.09412, 2017.

Dengyong Zhou and Bernhard Schölkopf. Regularization on discrete spaces. In Walter G. Kropatsch, Robert Sablatnig, and Allan Hanbury, editors, *Pattern Recognition*, pages 361–368, Berlin, Heidelberg, 2005. Springer Berlin Heidelberg. ISBN 978-3-540-31942-9.

Kaixiong Zhou, Xiao Huang, Daochen Zha, Rui Chen, Li Li, Soo-Hyun Choi, and Xia Hu. Dirichlet energy constrained learning for deep graph neural networks. *Advances in neural information processing systems*, 34:21834–21846, 2021.

A Additional Related Work

A.1 Learning under Label Noise

Beyond robust losses, additional families include sample relabelling [Arazo et al., 2019, Reed et al., 2015], two-network co-teaching and divide-and-mix approaches [Han et al., 2018, Li et al., 2020, Kim et al., 2023], and regularization through MixUp [Zhang et al., 2017]. Reweighting methods adapt sample importance via training dynamics or held-out probes [Liu and Tao, 2015, Pleiss et al., 2020]. Most of these were developed for image classification; their direct transfer to graph classification is non-obvious because graphs lack the strong feature regularities of images.

A.2 Graph Learning under Noise

Most graph-related noise work targets node classification, exploiting either homophily (RS-GNN [Dai et al., 2022], NRGNN [Dai et al., 2021], RTGNN [Yuan et al., 2023a], pairwise interactions [Du et al., 2023]) or self-supervision (pseudo-label denoising [Yuan et al., 2023b, Qian et al., 2023]). Edge/feature noise has also been studied, e.g. structural noise [Fox and Rajamanickam, 2019, Dai et al., 2022]. These methods typically assume connected nodes share labels, which is incompatible with our graph-classification setting where labels are graph-level. The seminal work of NT et al. [2019] treats graph classification under label noise via a surrogate loss; OMG [Yin et al., 2023] combines contrastive learning, MixUp, and curriculum filtering. Both are practical improvements but provide no spectral or energy-based explanation of *why* GNNs fit graph-level noise.

A.3 Dirichlet Energy and Spectral Properties of GNNs

NT and Maehara [2019] and Rusch et al. [2023] analyze GNNs as low-pass filters, with self-loops further shrinking eigenvalues. Cai and Wang [2020], Oono and Suzuki [2020] prove that GNN Dirichlet energy decays exponentially when the product of the largest singular value of the weight matrix and the largest Laplacian eigenvalue is below 1. Giovanni et al. [2023] disentangle smoothing/sharpening via positive/negative weight-eigenvalues. Zhou et al. [2021] (NeurIPS) uses energy regularization to prevent over-smoothing on *node* classification. Our setting is the opposite regime: too *much* energy under graph-level label noise.

Qian et al. [2025] introduce a structural-entropy viewpoint on over-smoothing for graph classification, showing that high-entropy regions are most damaged by over-smoothing. Their analysis covers the over-smoothing (low-energy) regime; we address the opposite regime where graph-level label noise drives *too much* within-graph energy. An explicit quantitative connection between structural entropy and noise-driven sharpening remains open and is a natural follow-up direction.

A.4 Lipschitz/Stability Analyses

Chuang and Jegelka [2022] bound GIN outputs via the Tree Mover's Distance, and Davidson and Dym [2025] extend uniform-Lipschitz analyses to MPNNs through Hölder continuity of aggregation/combination/readout. Juvina et al. [2024] derive optimal Lipschitz constants $\vartheta = \phi(\lambda_K)$ for graph convolutions under non-negative weights and ReLU. Gama et al. [2019] show output stability under graph-shift-operator perturbations. These results characterize stability w.r.t. inputs; we instead study stability against *label* noise, mediated by the within-graph energy of learned representations.

B Proofs and Extended Theory

B.1 Proof of Proposition 1

Let $\Lambda^i = \{\lambda_u^i\}_{u=1}^{N^i}$ be the spectrum of the per-graph combinatorial Laplacian $\mathbf{L}^i = \mathbf{D}^i - \mathbf{A}^i$. Define the block-diagonal Laplacian

$$\bar{\mathbf{L}} = \operatorname{blkdiag}(\mathbf{L}^1, \ldots, \mathbf{L}^n), \qquad \det(\bar{\mathbf{L}} - \lambda \mathbf{I}_{N_{\text{tot}}}) = \prod_{i=1}^{n} \det(\mathbf{L}^i - \lambda \mathbf{I}_{N^i}), \tag{12}$$

so $\bar{\Lambda} = \bigcup_i \Lambda^i$. Let ψ_j^i be the j-th eigenvector of $\mathbf{L}^i$ at eigenvalue λ_j^i. The vector

$$\bar{\psi}_u = \big(0, \ldots, 0, \psi_j^i, 0, \ldots, 0\big)^\top \in \mathbb{R}^{N_{\text{tot}}} \tag{13}$$

(zero on every block other than i) satisfies $\bar{\mathbf{L}}\bar{\psi}_u = \lambda_j^i \bar{\psi}_u$. With $\bar{\mathbf{Z}} = [\mathbf{Z}^1 \| \ldots \| \mathbf{Z}^n]$ (vertical concatenation), the combinatorial Dirichlet energy of the block-diagonal graph satisfies

$$E^{\text{dir}}(\bar{\mathbf{Z}}) = \operatorname{tr}(\bar{\mathbf{Z}}^\top \bar{\mathbf{L}} \bar{\mathbf{Z}}) = \sum_{r=1}^{m'} \sum_{u=1}^{N_{\text{tot}}} \bar{\lambda}_u (\bar{\psi}_u^\top \bar{\mathbf{Z}}_{:,r})^2 = \sum_i \sum_{r,j} \lambda_j^i (\psi_j^{i\top} \mathbf{Z}^i_{:,r})^2 = \sum_i E^{\text{dir}}(\mathbf{Z}^i), \tag{14}$$

where the second equality uses the spectral expansion in the orthonormal basis $\{\bar{\psi}_u\}$ and the third uses the block-supported structure of each $\bar{\psi}_u$. Hence $E^{\text{dir}}_{\text{set}}(\mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_i E^{\text{dir}}(\mathbf{Z}^i) = \frac{1}{|\mathcal{D}|} E^{\text{dir}}(\bar{\mathbf{Z}})$. $\square$

B.2 Proof of Lemma 1 (Graph Poincaré Inequality)

The combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ is real symmetric and positive semi-definite. Because $\mathcal{G}$ is connected, $\ker(\mathbf{L}) = \operatorname{span}(\mathbf{1})$, and the eigenvalues are $0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_N$. By the Courant–Fischer min–max characterization,

$$\lambda_2(\mathcal{G}) = \min_{\substack{\mathbf{x} \perp \mathbf{1} \\ \mathbf{x} \neq 0}} \frac{\mathbf{x}^\top \mathbf{L} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}. \tag{15}$$

Fix any feature dimension r and let $\mathbf{z} = \mathbf{Z}_{:,r} \in \mathbb{R}^N$ with $\bar{z} = \frac{1}{N} \sum_v z_v$. The centered vector $\mathbf{z} - \bar{z}\mathbf{1}$ is orthogonal to $\mathbf{1}$, so

$$\lambda_2(\mathcal{G}) \, \|\mathbf{z} - \bar{z}\mathbf{1}\|_2^2 \ \leq\ (\mathbf{z} - \bar{z}\mathbf{1})^\top \mathbf{L} (\mathbf{z} - \bar{z}\mathbf{1}) = \mathbf{z}^\top \mathbf{L} \mathbf{z} = \sum_{(u,v) \in \mathcal{E}} (z_u - z_v)^2, \tag{16}$$

using $\mathbf{L1} = \mathbf{0}$. The right-hand side is the per-feature-dimension contribution to $E^{\text{dir}}(\mathbf{Z})$ in (1). Summing over $r = 1, \ldots, m'$ yields

$$\sum_{v=1}^{N} \|\mathbf{Z}_v - \bar{\mathbf{z}}\|_2^2 = \sum_{r=1}^{m'} \|\mathbf{z}^{(r)} - \bar{z}^{(r)}\mathbf{1}\|_2^2 \leq \frac{1}{\lambda_2(\mathcal{G})} \sum_{r=1}^{m'} \mathbf{z}^{(r)\top} \mathbf{L} \mathbf{z}^{(r)} = \frac{E^{\text{dir}}(\mathbf{Z})}{\lambda_2(\mathcal{G})}, \tag{17}$$

which is (4). The constant $1/\lambda_2$ is sharp, attained by the Fiedler vector ψ_2. The corresponding bound for the normalized Dirichlet energy $E^{\text{dir}}_{\text{sym}}(\mathbf{Z}) = \operatorname{tr}(\mathbf{Z}^\top \boldsymbol{\Delta} \mathbf{Z})$ is obtained by the change of variable $\mathbf{y}_v = \mathbf{Z}_v / \sqrt{d_v}$ and applying the same argument with $\lambda_2(\mathbf{L})$ in the bound. $\square$

B.3 Proof of Lemma 2 (Lower Lipschitz Bound for a GIN)

The block under analysis is one GIN aggregation followed by a two-layer MLP combine,

$$h_\theta(\mathbf{X}) = \sigma_2\big(\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\, \mathbf{W}_2\big), \tag{18}$$

acting on $\mathbf{X} \in \mathbb{R}^{N \times m}$ with $\mathbf{M} \in \mathbb{R}^{N \times N}$ mixing rows, and $\mathbf{W}_1 \in \mathbb{R}^{m \times m'}$, $\mathbf{W}_2 \in \mathbb{R}^{m' \times m''}$ acting on the feature dimension.

Global lower-Lipschitz of leaky-ReLU. For scalar $a, b \in \mathbb{R}$, $|\mathrm{LReLU}_\alpha(a) - \mathrm{LReLU}_\alpha(b)| \geq \alpha |a - b|$, since the function has slope $\geq \alpha > 0$ everywhere. Applied element-wise to a matrix $\mathbf{Y}$ and a perturbation $\mathbf{Y} + \Delta\mathbf{Y}$ this gives

$$\|\sigma_k(\mathbf{Y} + \Delta\mathbf{Y}) - \sigma_k(\mathbf{Y})\|_F \ \geq\ \alpha \, \|\Delta\mathbf{Y}\|_F, \qquad k = 1, 2. \tag{19}$$

This is a *global* statement, not a local one — no strict-margin or single-region assumption is needed for leaky-ReLU.

Lower bounds for right- and left-multiplication. Using $\operatorname{vec}(\mathbf{AYB}) = (\mathbf{B}^\top \otimes \mathbf{A}) \operatorname{vec}(\mathbf{Y})$ and $\sigma_{\min}(\mathbf{P} \otimes \mathbf{Q}) = \sigma_{\min}(\mathbf{P}) \sigma_{\min}(\mathbf{Q})$, for any matrix $\mathbf{Z}$,

$$\|\mathbf{Z}\mathbf{W}_2\|_F = \|(\mathbf{W}_2^\top \otimes \mathbf{I}) \operatorname{vec}(\mathbf{Z})\|_2 \geq \sigma_{\min}(\mathbf{W}_2) \|\mathbf{Z}\|_F \geq m_2 \|\mathbf{Z}\|_F, \tag{20}$$

$$\|\mathbf{M}\mathbf{Z}\mathbf{W}_1\|_F = \|(\mathbf{W}_1^\top \otimes \mathbf{M}) \operatorname{vec}(\mathbf{Z})\|_2 \geq \sigma_{\min}(\mathbf{W}_1) \sigma_{\min}(\mathbf{M}) \|\mathbf{Z}\|_F \geq m_1 \kappa \|\mathbf{Z}\|_F. \tag{21}$$

The widening regime $m \leq m' \leq m''$ together with $\sigma_{\min}(\mathbf{W}_k) > 0$, $\sigma_{\min}(\mathbf{M}) > 0$ ensures that each Kronecker factor has full column rank, so the lower-bound inequalities above are non-trivial.

Chain. Apply (19) for σ_2, then right-multiplication by $\mathbf{W}_2$, then (19) for σ_1, then $\mathbf{M}\cdot\mathbf{W}_1$:

$$\begin{aligned}\|h_\theta(\mathbf{X}+\Delta\mathbf{X}) - h_\theta(\mathbf{X})\|_F &\geq \alpha \,\|\sigma_1(\mathbf{M}(\mathbf{X}+\Delta\mathbf{X})\mathbf{W}_1)\mathbf{W}_2 - \sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2\|_F \\ &\geq \alpha m_2 \,\|\sigma_1(\mathbf{M}(\mathbf{X}+\Delta\mathbf{X})\mathbf{W}_1) - \sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\|_F \\ &\geq \alpha^2 m_2 \,\|\mathbf{M}\Delta\mathbf{X}\mathbf{W}_1\|_F \\ &\geq \alpha^2 m_1 m_2 \kappa \,\|\Delta\mathbf{X}\|_F = c_{\text{lower}} \,\|\Delta\mathbf{X}\|_F. \end{aligned} \tag{22}$$

Equivalent vectorized Jacobian view. Writing the Jacobian in vec form, $\mathbf{J} = \mathbf{D}_2(\mathbf{W}_2^\top \otimes \mathbf{I}_N)\mathbf{D}_1(\mathbf{W}_1^\top \otimes \mathbf{M})$, with $\mathbf{D}_k \succeq \alpha\mathbf{I}$ for leaky-ReLU. Using $\sigma_{\min}(\mathbf{P}\otimes\mathbf{Q}) = \sigma_{\min}(\mathbf{P})\sigma_{\min}(\mathbf{Q})$ and the singular-value chain $\sigma_{\min}(\mathbf{ABCD}) \geq \prod \sigma_{\min}$ (valid under full column rank of each factor) recovers $\sigma_{\min}(\mathbf{J}) \geq \alpha^2 m_1 m_2 \kappa$. The element-wise chain above is what turns this Jacobian bound into the global statement (5) for leaky-ReLU.

Two-layer GIN. For $\sigma_2(\mathbf{M}\sigma_1(\mathbf{M}\mathbf{X}\mathbf{W}_1)\mathbf{W}_2)$, an additional left-multiplication by $\mathbf{M}$ enters the chain, contributing an extra $\sigma_{\min}(\mathbf{M}) \geq \kappa$ and giving $c_{\text{lower}} = \alpha^2 m_1 m_2 \kappa^2$.

Ordinary ReLU. The element-wise lower-Lipschitz bound (19) fails globally for ReLU: on any segment that crosses the origin in some coordinate, $|\mathrm{ReLU}(a) - \mathrm{ReLU}(b)| < |a-b|$ is possible. The chain therefore only works on regions where every coordinate of every preactivation is strictly positive along $\gamma(t) = \mathbf{X} + t\Delta\mathbf{X}$. On such a region, σ_k acts as the identity (slope 1), so the chain gives $c_{\text{lower}} = m_1 m_2 \kappa$ (the formula in (5) with $\alpha = 1$). $\square$

B.4 Proof of Theorem 1 (Noise Forces Energy Growth)

Let $\mathbf{Z}^i, \mathbf{Z}^j \in \mathbb{R}^{N\times m'}$ be the post-message-passing per-node representations of $\mathcal{G}^i$ and $\mathcal{G}^j$ as defined in Section 3 (we assume $N^i = N^j = N$; the unequal-N case is treated below). Let $\mathbf{g}_i = \frac{1}{N}\sum_v \mathbf{Z}^i_v \in \mathbb{R}^{m'}$ be the mean-pooled graph descriptor, and let $\widetilde{\mathbf{Z}}^i = \mathbf{1}_N \mathbf{g}_i^\top$ denote its replication to a constant per-node signal. By the Frobenius-norm triangle inequality,

$$\|\mathbf{Z}^i - \mathbf{Z}^j\|_F \;\leq\; \|\mathbf{Z}^i - \widetilde{\mathbf{Z}}^i\|_F + \|\widetilde{\mathbf{Z}}^i - \widetilde{\mathbf{Z}}^j\|_F + \|\widetilde{\mathbf{Z}}^j - \mathbf{Z}^j\|_F. \tag{23}$$

Middle term. Because both $\widetilde{\mathbf{Z}}^i, \widetilde{\mathbf{Z}}^j$ are constant signals of size $N \times m'$,

$$\|\widetilde{\mathbf{Z}}^i - \widetilde{\mathbf{Z}}^j\|_F^2 = \sum_{v=1}^{N} \|\mathbf{g}_i - \mathbf{g}_j\|_2^2 = N\,\|\mathbf{g}_i - \mathbf{g}_j\|_2^2, \tag{24}$$

so $\|\widetilde{\mathbf{Z}}^i - \widetilde{\mathbf{Z}}^j\|_F = \sqrt{N}\|\mathbf{g}_i - \mathbf{g}_j\|_2$.

Variance terms. Lemma 1 applied to $\mathbf{Z}^i$ gives

$$\|\mathbf{Z}^i - \widetilde{\mathbf{Z}}^i\|_F^2 = \sum_v \|\mathbf{Z}^i_v - \mathbf{g}_i\|_2^2 \;\leq\; \frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}, \tag{25}$$

and analogously for $\mathcal{G}^j$. Taking square roots and combining,

$$\|\mathbf{Z}^i - \mathbf{Z}^j\|_F \;\leq\; \sqrt{N}\,\|\mathbf{g}_i - \mathbf{g}_j\|_2 + \sqrt{\tfrac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\tfrac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}}, \tag{26}$$

which is (6).

Lower bound from the noisy training objective. The graph-level prediction is $\hat{\mathbf{y}}_i = f_\theta(\mathbf{Z}^i)$, where f_θ is the permutation-invariant readout from Section 3; we assume f_θ is L-Lipschitz in the Frobenius norm (mean- or sum-pooling followed by an MLP with bounded weights). At convergence, the noisy training objective enforces $\hat{\mathbf{y}}_i$ to lie near a one-hot vector at c_i and $\hat{\mathbf{y}}_j$ near a one-hot at $c_j \neq c_i$, so $\|\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j\|_2 \geq \Delta_{\text{out}}$ for some $\Delta_{\text{out}} > 0$. By the readout's upper-Lipschitz property,

$$\|\mathbf{Z}^i - \mathbf{Z}^j\|_F \;\geq\; L^{-1}\,\|\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j\|_2 \;\geq\; L^{-1}\Delta_{\text{out}} =: \Theta_{\text{noise}}. \tag{27}$$

The role of Lemma 2 is to make this lower bound non-vacuous: $c_{\text{lower}} > 0$ ensures that the encoder has a positive lower-Lipschitz constant at convergence, so the noisy-label constraint cannot be satisfied by collapsing $(\mathbf{X}^i, \mathbf{A}^i)$ and $(\mathbf{X}^j, \mathbf{A}^j)$ to the same internal representation; equivalently, the encoder remains in a regularity regime where $\|\mathbf{Z}^i - \mathbf{Z}^j\|_F$ is a meaningful quantity.

Combining upper and lower bounds. Under the structural-similarity assumption $\|\mathbf{g}_i - \mathbf{g}_j\|_2 \leq \varepsilon$, substituting $\Theta_{\text{noise}} \leq \|\mathbf{Z}^i - \mathbf{Z}^j\|_F$ into (6) and rearranging,

$$\sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^i)}{\lambda_2(\mathcal{G}^i)}} + \sqrt{\frac{E^{\text{dir}}(\mathbf{Z}^j)}{\lambda_2(\mathcal{G}^j)}} \ \geq\ \Theta_{\text{noise}} - \sqrt{N}\,\varepsilon, \tag{28}$$

which is (7). $\square$

Unequal graph sizes. When $N^i \neq N^j$, set $N = \max(N^i, N^j)$ and pad the smaller graph by appending $N - N^{\min}$ rows equal to its mean $\mathbf{g}_\bullet$, with the corresponding padded vertices isolated (no edges). Let $(\mathbf{Z}^\bullet)^{\text{pad}}$ and $(\widetilde{\mathbf{Z}}^\bullet)^{\text{pad}}$ denote the padded representation and its constant-signal replication. Two facts make this padding harmless: (i) appending $\mathbf{g}_\bullet$-valued rows leaves the graph mean unchanged, and (ii) isolated padded vertices contribute nothing to the Dirichlet energy, so $E^{\text{dir}}((\mathbf{Z}^\bullet)^{\text{pad}}) = E^{\text{dir}}(\mathbf{Z}^\bullet)$. The within-graph variance $\|(\mathbf{Z}^\bullet)^{\text{pad}} - (\widetilde{\mathbf{Z}}^\bullet)^{\text{pad}}\|_F^2 = \sum_{v=1}^{N^{\min}} \|\mathbf{Z}^\bullet_v - \mathbf{g}_\bullet\|^2$ is also preserved, so Lemma 1 applies unchanged. The middle term becomes $\|(\widetilde{\mathbf{Z}}^i)^{\text{pad}} - (\widetilde{\mathbf{Z}}^j)^{\text{pad}}\|_F = \sqrt{N}\|\mathbf{g}_i - \mathbf{g}_j\|_2$, and the readout's lower bound is preserved up to a fixed pooling constant. Hence (6) and (7) hold with $N \to \max(N^i, N^j)$.

B.5 Why a Decrease of $E^{\text{dir}}_{\text{set}}$ Is Non-Trivial in Graph Classification

Three consecutive observations together justify the term "non-trivial":

(i) Noise is injected at the graph-label level, not at the node-feature level. Therefore, the increase of $E^{\text{dir}}(\mathbf{Z}^i)$ during noise fitting cannot be a re-encoding of the noise itself; it must be a behavioral response of the network through training dynamics.

(ii) For a randomly-initialized network, $E^{\text{dir}}_{\text{set}}$ is independent of class labels by construction. Any correlation between $E^{\text{dir}}_{\text{set}}$ and labels emerges from the optimization process, not from the data.

(iii) By Theorem 1, fitting incorrect graph-level labels mathematically forces within-graph energy to grow. Constraining $E^{\text{dir}}_{\text{set}}$ therefore directly opposes the noise-fitting path, an alignment that does *not* hold for node classification, where label noise corrupts the same nodes whose smoothness the energy measures, and where decreasing energy can erase the discriminative signal.

These three points collectively make the connection between energy reduction and noise robustness in graph classification a non-trivial property. The analogous statement in node classification reduces to "the model fits homophily, and homophily is what we measured."

C Experimental Setting

C.1 Datasets

ogbg-ppa [Szklarczyk et al., 2018a]: 158,100 protein-association graphs across 37 taxonomic groups, derived from STRING [Szklarczyk et al., 2018b]; the standard OGB graph-property-prediction benchmark. Modified PPA-30%-6 takes 30% of training graphs from 6 selected classes, used as a controlled testbed. **PROTEINS** [Borgwardt et al., 2005]: 1,113 protein graphs; 2 classes; mean 39 nodes/graph. **ENZYMES** [Borgwardt et al., 2005]: 600 enzyme graphs across 6 EC top-level classes. **MSRC-21** [Neumann et al., 2016]: 563 image-segmentation graphs, 20 classes. **MUTAG** [Kriege and Mutzel, 2012]: 188 molecular graphs, 2 classes. **IMDB-BINARY** [Yanardag and Vishwanathan, 2015b]: 1,000 social graphs, 2 classes. **REDDIT-MULTI-12K** [Yanardag and Vishwanathan, 2015b]: 11,929 thread-graphs, 11 classes. **MNIST-graph**: 55,000 superpixel graphs over the MNIST images, 10 classes.

C.2 Synthetic Datasets for Failure-Mode F2

We sample average graph orders $\bar{N} \in \{5, 10, \ldots, 60\}$, drawing actual node counts from $\text{Poisson}(\bar{N})$. Each dataset has 6 classes with 1,400 graphs/class, fixed average degree 2, edges sampled uniformly. Node features are $\mathcal{N}(\mu_c, 1.5)$ with $\mu_c = f(\text{class})$. Symmetric noise rate is fixed at 35%.

C.3 Architectures and Hyperparameters

Unless otherwise stated, we use a 5-layer GIN [Xu et al., 2019] or GCN [Kipf and Welling, 2017] with 300 hidden units, Adam ($\eta = 10^{-3}$, weight decay 10^{-4}), batch size 32, 200–1000 epochs, $\eta_u = 1$ for GCOD's outlier parameter. Compute: NVIDIA RTX A6000.

C.4 Spectral Bias Implementation Details (CE+W2)

For each layer l, after a gradient update, decompose $\mathbf{W}_l^2 = \mathbf{\Phi}_l^2 \boldsymbol{\mu}_l^2 (\mathbf{\Phi}_l^2)^{-1}$ and replace by $\mathbf{\Phi}_l^2 [\boldsymbol{\mu}_l^2]^+ (\mathbf{\Phi}_l^2)^{-1}$, where $[\cdot]^+$ is element-wise ReLU on the eigenvalues. The projection is detached from the gradient graph. We apply it to the post-aggregation matrix $\mathbf{W}^2$ of every layer; applying it also to $\mathbf{W}^1$ marginally improves smoothing but degrades stability. Convergence variance under this scheme is shown in Fig. 8.

C.5 Class-Specific and Fixed Bounds for $\mathcal{L}_{DE}$

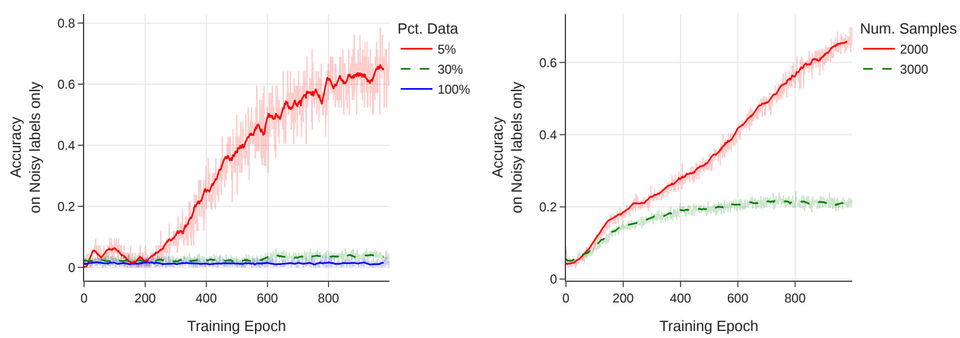
Class-specific: after every epoch, set $U_c = |\mathcal{D}_c^{\text{val}}|^{-1} \sum_{i \in \mathcal{D}_c^{\text{val}}} E^{\text{dir}}(\mathbf{Z}^i)$. The validation set must be clean to keep U_c class-specific. **Fixed:** $U_c = U$ globally. We progressively decrease U from a large initial value until accuracy under clean labels begins to drop; the final U is the largest value that achieves a noticeable energy ceiling without over-smoothing.

C.6 Design of GCOD

We adopt the centroid-based soft labels of Wani et al. [2023]: normalize $\phi(\mathbf{z}_i) = \mathbf{z}_i / \|\mathbf{z}_i\|$; per-class centroid $\boldsymbol{\phi}_c = \bar{\mathbf{z}}_c / \|\bar{\mathbf{z}}_c\|$; soft target $\tilde{\mathbf{y}}_i = [\max(\phi(\mathbf{z}_i)\boldsymbol{\phi}_c^\top, 0)\mathbf{y}_i]$. Equations (9)–(11) are then optimized as in the main text. The training-accuracy feedback in $\mathcal{L}_1$ and $\mathcal{L}_3$ is critical: without it, $\mathbf{u}$ dominates early in training and discounts samples before genuine patterns are learned (Fig. 7).

D Failure-Mode Experiments (Extended)

Additional plots for the F1–F3 failure modes are provided here for completeness, including effect-of-dataset-size for both PPA and synthetic datasets.



(a) (b)

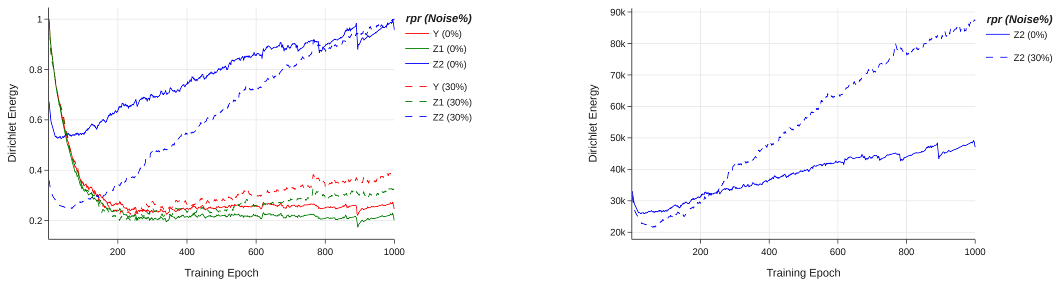
Figure 3: Effect of training-set size on noise memorization (training accuracy on noisy labels only). (a) PPA subsampling at 20% symmetric noise. (b) Synthetic graphs of order 7, 6 classes, 35% noise.

E Dirichlet-Energy Amplification in High-Frequency Components

We use the HLFF-GNN framework [Xu et al., 2024] (implemented as `FGRLConv`) to decompose representations into low-frequency Z_1, high-frequency Z_2, and residual Y. Under the composite loss

$$\mathcal{L} = \|Y - X\|_F^2 + \lambda \, \mathrm{tr}(Z_1^\top \mathbf{\Delta} Z_1) + \alpha \, \mathrm{tr}(Z_2^\top (\mathbf{I} - \mathbf{\Delta}) Z_2) + \beta(\|Y^\top Z_1\|_F^2 + \|Y^\top Z_2\|_F^2), \quad (29)$$

the spectral form of the high-frequency energy is $E^{\text{dir}}(Z_2) = \sum_{r,u} \lambda_u (\boldsymbol{\psi}_u^\top Z_{2,r})^2$, in which large λ_u amplify any noise-driven content. On ENZYMES with 30% symmetric noise, we observe a sharp monotone increase in $E^{\text{dir}}(Z_2)$, while $E^{\text{dir}}(Z_1)$ and $E^{\text{dir}}(Y)$ remain stable, consistent with Theorem 1.



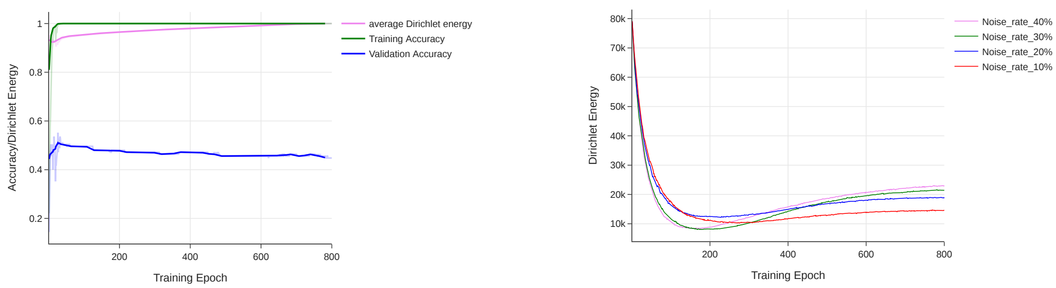
(a) Normalized $E^{\text{dir}}(Y, Z_1, Z_2)$. (b) $E^{\text{dir}}(Z_2)$ alone.

Figure 4: HLFF-GNN frequency decomposition on ENZYMES. Solid lines: clean labels. Dashed: 30% symmetric noise. The high-frequency component Z_2 is the only one whose energy diverges under noise, supporting the theoretical prediction.

F Additional Empirical Studies on Energy Behavior

Energy under clean overfitting vs. noise. On ENZYMES without regularization, a deliberately overfit GIN exhibits rising training accuracy and rising $E^{\text{dir}}_{\text{set}}$, confirming that energy tracks overfitting in general. However, the rise under noise is markedly steeper.

Energy scales with noise rate. On the controlled PPA subset, we sweep symmetric noise $\in \{10, 20, 30, 40, 60\}\%$. All curves follow the same U-shape (initial decrease, then rise), and the asymptotic energy increases monotonically with noise rate. This monotonicity is what makes $E^{\text{dir}}_{\text{set}}$ a diagnostic.



(a) Clean overfitting on ENZYMES. (b) $E^{\text{dir}}_{\text{set}}$ vs. noise rate on PPA.

Figure 5: (a) Even without label noise, energy rises during overfitting; the surge under noise is much steeper. (b) Higher noise rates yield monotonically higher final energy.

G Additional Tables and Figures

H Full ogbg-ppa: Do-No-Harm Check

The standard ogbg-ppa benchmark has 37 classes and ∼158,000 graphs of mean order ≈240, so a 5-layer GIN with 300 hidden units is structurally *under-parameterized* for the task. We trained CE, $\mathcal{L}_{DE}$ (Fixed), CE+W2 and GCOD on the full benchmark with 0%, 20% and 40% symmetric noise. Two qualitative observations are robust across runs:

1. *Training accuracy on the noisy-labeled subset stays close to chance for all methods, including standard CE.* The under-parameterized model never reaches a regime where it can memorize the corrupted graph-level labels, consistent with our failure-mode analysis of Section 4: F1 (number

Table 5: Sensitivity of GCOD to the learning rate η_u of the outlier parameter $\mathbf{u}$, under 40% asymmetric noise.

Dataset	$\eta_u = 1$	$\eta_u = 0.1$	% Change
PROTEINS	76.19	75.89	-0.39%
MNIST	72.61	72.48	-0.18%
ENZYMES	69.81	68.33	-2.12%
IMDB-B	72.89	71.94	-1.30%
MUTAG	93.19	92.61	-0.62%
REDDIT	47.94	48.08	$+0.29\%$
MSRC-21	95.57	95.45	-0.13%

Table 6: Performance of CE vs. GCOD with 20% symmetric label noise. Last column reports the absolute gain (GCOD$-$CE) on test accuracy.

Dataset	**Metric**	**0% CE**	**20% CE**	**20% GCOD**	**20%** (GCOD$-$CE)
MNIST	Best	72.69	66.30	71.26	+4.96
	Last	69.80	53.64	67.88	+14.24
	Diff	2.89	12.66	3.38	
ENZYMES	Best	73.33	64.16	68.69	+4.53
	Last	65.78	57.50	62.54	+5.04
	Diff	7.55	6.66	6.15	
MSRC-21	Best	96.69	90.26	94.69	+4.43
	Last	93.80	79.64	90.26	+10.62
	Diff	2.89	10.62	4.43	
PROTEINS	Best	81.16	76.23	79.38	+3.15
	Last	79.18	62.32	78.12	+15.80
	Diff	1.98	13.91	1.26	
MUTAG	Best	94.73	89.47	90.01	+0.54
	Last	84.21	68.42	86.84	+18.42
	Diff	10.52	21.05	3.17	
IMDB-B	Best	76.50	75.00	75.40	+0.40
	Last	71.60	71.00	73.50	+2.50
	Diff	4.90	4.00	1.90	
REDDIT	Best	48.15	45.05	44.98	-0.07
	Last	46.01	44.67	44.89	+0.22
	Diff	2.14	0.38	0.09	

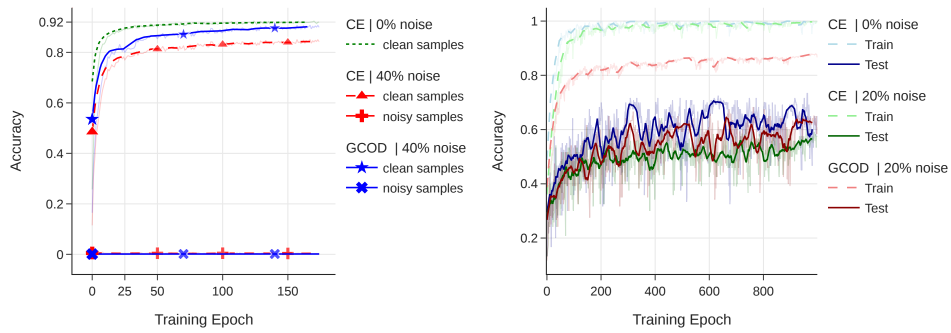
of classes), F2 (graph order) and F3 (training-set size) all push the full PPA out of the noise-fitting regime.

2. *Test accuracy of our three methods is statistically indistinguishable from CE* at every noise rate. None of CE+W2, $\mathcal{L}_{DE}$, or GCOD perturbs the underlying training dynamics enough to harm clean generalization, since their corrective signals (negative-eigenvalue projection, energy upper bound, outlier discounting) are only triggered by the high-frequency surge that does not occur on the full PPA.

This is by design: the methods are conditional regularizers, not unconditional smoothers. The take-away is that they impose *no measurable overhead in accuracy* when the underlying network is already noise-robust, while providing substantial protection when over-parameterization, low graph order, or small training sets push the network into the noise-fitting regime (Section 8, controlled PPA-30%-6 subset). A fair generality claim therefore rests on the seven additional benchmarks in Tables 2 and 6, with the full PPA serving exclusively as a do-no-harm sanity check.

I Cross-Setting Transfer: Cora Node Classification

To assess whether the energy perspective captures a property of GNN training under label noise that is more general than the graph-classification setting it was designed for, we apply CE+W2 and GCOD *unchanged* to node classification on Cora. We use both GAT and GCN backbones with the standard public split, the transductive setting, and compare against 13 baselines: standard cross-entropy (CE; column "standard"), UnionNet, GraphCleaner, GNNGuard, GNN-Cleaner, PI-GNN, NRGNN [Dai



(a) (b)

Figure 6: (a) Train accuracy on known noisy and clean PPA subsets (4-class, 40% symmetric noise) for GCN with CE: the model never separates noisy from clean training samples without a robust loss. (b) Train/test accuracy on ENZYMES with GIN under clean and 20% symmetric noise.

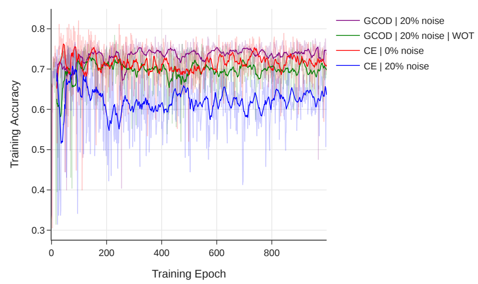


Figure 7: Ablation on the training-accuracy feedback in GCOD. Without scaling u by a_{train} ("GCOD WOT"), the outlier parameter dominates early and harms generalization; the full GCOD activates outlier-discounting gradually.

et al., 2021], RTGNN [Yuan et al., 2023a], Community-Defense, ERASE, and CR-GNN. We evaluate four label-noise regimes: clean (no noise), symmetric uniform 20%, symmetric random 20%, and pairwise flip 20%. Each entry is mean±std over 3 independent runs.

Findings. Tables 7 and 8 show that on every backbone-noise combination, at least one of CE+W2 or GCOD ranks in the top four among the compared methods, with most placements in the top three, despite neither using neighbor labels, homophily, contrastive losses, pseudo-labels, nor any other node-specific machinery. Highlights: GCOD reaches 0.787 under uniform-20% on GAT (#2 of 10), 0.792 on clean GCN (#3 of 13), and 0.769 on flip-20% GCN (#3 of 13); CE+W2 reaches 0.816 on clean GAT (#2 of 13) and 0.774 on random-20% GCN (#2 of 13). The single combination where neither of our methods cracks the top three is Cora-GAT random-20%, where GCOD is #4 of 10. The only baseline that consistently outperforms both is UnionNet, a node-classification-specific method that exploits label-propagation priors unavailable in the graph-classification setting our methods were designed for.

Interpretation, consistent with Theorem 1. The spectral mechanism by which noise fitting forces within-graph energy to grow is not specific to graph classification: in node classification, individual labels are also embedded in node-level representations whose Dirichlet energy reflects local smoothness, and constraining that smoothness penalizes the same noise-fitting path. The transfer is therefore not a coincidence but a corollary of the energy-based framing. Additional comparisons on `dblp` (GCN clean) and `amazon-computers` (GCN clean), available in the released code repository, show CE+W2 reaching 0.866 on `dblp` (#3 of 13) and competitive accuracy on `amazon-computers`, further supporting the cross-setting generality.

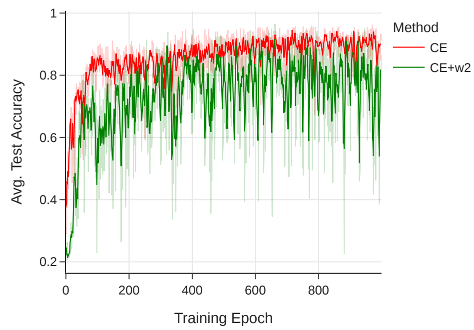


Figure 8: Test accuracy on PPA-30%-6 for CE and CE+W2. CE+W2 reaches comparable peak accuracy but exhibits unstable convergence due to the per-epoch eigendecomposition.

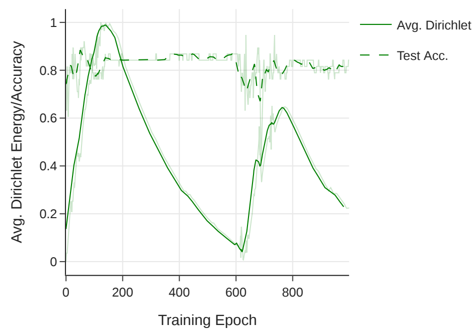


Figure 9: Average test accuracy and average within-graph $E_{\text{set}}^{\text{dir}}$ on MUTAG (0% noise) with GIN: classification accuracy improves while energy decays, the clean-data signature of a healthy GNN.

Table 7: Cora node classification, GAT backbone. Each cell is test accuracy (mean over 3 runs; std deviations available in the released code repository). Best per column is bold; our methods (CE+W2 and GCOD) are highlighted in gray. "–" denotes settings not run for the given baseline.

Method	**Clean**	**Flip 20%**	**Random 20%**	**Uniform 20%**
UnionNet	**0.8287**	**0.7997**	**0.8183**	**0.8060**
CE+W2 (positive_eig.)	0.8157	0.7660	0.7603	0.7483
GCOD (gcod)	0.8120	0.7607	0.7733	0.7870
GraphCleaner	0.8023	0.7920	0.7877	0.7690
GNNGuard	0.7983	0.7527	0.7787	0.7460
Community-Defense	0.7953	0.7440	0.7680	0.7390
CE (standard)	0.7950	0.7440	0.7693	0.7437
PI-GNN	0.7927	0.7303	0.7663	0.7390
RTGNN	0.7807	0.7247	0.7553	0.7203
GNN-Cleaner	0.7790	0.6917	–	–
ERASE	0.7710	0.7350	0.7703	0.7483
CR-GNN	0.7570	0.6803	–	–
NRGNN	0.6887	0.6207	–	–

Table 8: Cora node classification, GCN backbone. Each cell is test accuracy (mean over 3 runs). Best per column is bold; our methods are highlighted in gray.

Method	**Clean**	**Flip 20%**	**Random 20%**
UnionNet	**0.8060**	**0.7960**	**0.8040**
NRGNN	0.7937	0.6790	0.7270
GCOD (gcod)	0.7920	0.7687	0.7613
Community-Defense	0.7903	0.7213	0.7573
CE+W2 (positive_eig.)	0.7880	0.7543	0.7743
GraphCleaner	0.7860	0.7797	0.7700
GNN-Cleaner	0.7760	0.6650	0.7050
GNNGuard	0.7700	0.7073	0.7130
CE (standard)	0.7700	0.7073	0.7130
PI-GNN	0.7620	0.6830	0.7087
RTGNN	0.7543	0.7033	0.7017
CR-GNN	0.7320	0.6100	0.6513
ERASE	0.6957	0.6597	0.6897

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state precisely the claims supported by the theoretical results in Section 6 and the experiments in Section 8.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: A dedicated paragraph in the Conclusion discusses the absence of a closed-form noise-energy characterization, reliance on synthetic noise, and the lack of standardized real-world noisy graph-classification benchmarks.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Proposition 1, Lemmas 1–2, and Theorem 1 state their assumptions explicitly in the main text, and full proofs are provided in Appendix B.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper?

   Answer: [Yes]

   Justification: Appendix C provides datasets, splits, hyperparameters, architectures, and the implementation details of all three methods. Anonymized code is provided in the supplemental material.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: [Yes]

   Justification: All datasets are public benchmarks; an anonymous code repository accompanies the submission.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: [Yes]

   Justification: Section 8 and Appendix C report all training and evaluation details.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: All main results report mean$\pm$std over 4 independent runs with different seeds.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources?

   Answer: [Yes]

   Justification: Appendix C reports the use of NVIDIA RTX A6000 GPUs and Table 4 reports relative training-time overheads.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

   Answer: [Yes]

   Justification: The work uses public benchmarks, does not involve human subjects, and the broader-impact paragraph discusses application contexts.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Discussed in the Conclusion. The work primarily benefits scientific applications (chemistry, biology) where label noise is systematic.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

    Answer: [N/A]

    Justification: The paper does not release new datasets or pretrained models with misuse risk.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: All datasets are cited; OGB, TU-Datasets and PyTorch-Geometric implementations are used under their respective licenses.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The released code provides documentation for the proposed methods.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: No human-subject or crowdsourcing experiments.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human subjects.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

    Answer: [N/A]

    Justification: LLMs are not part of the core methods.